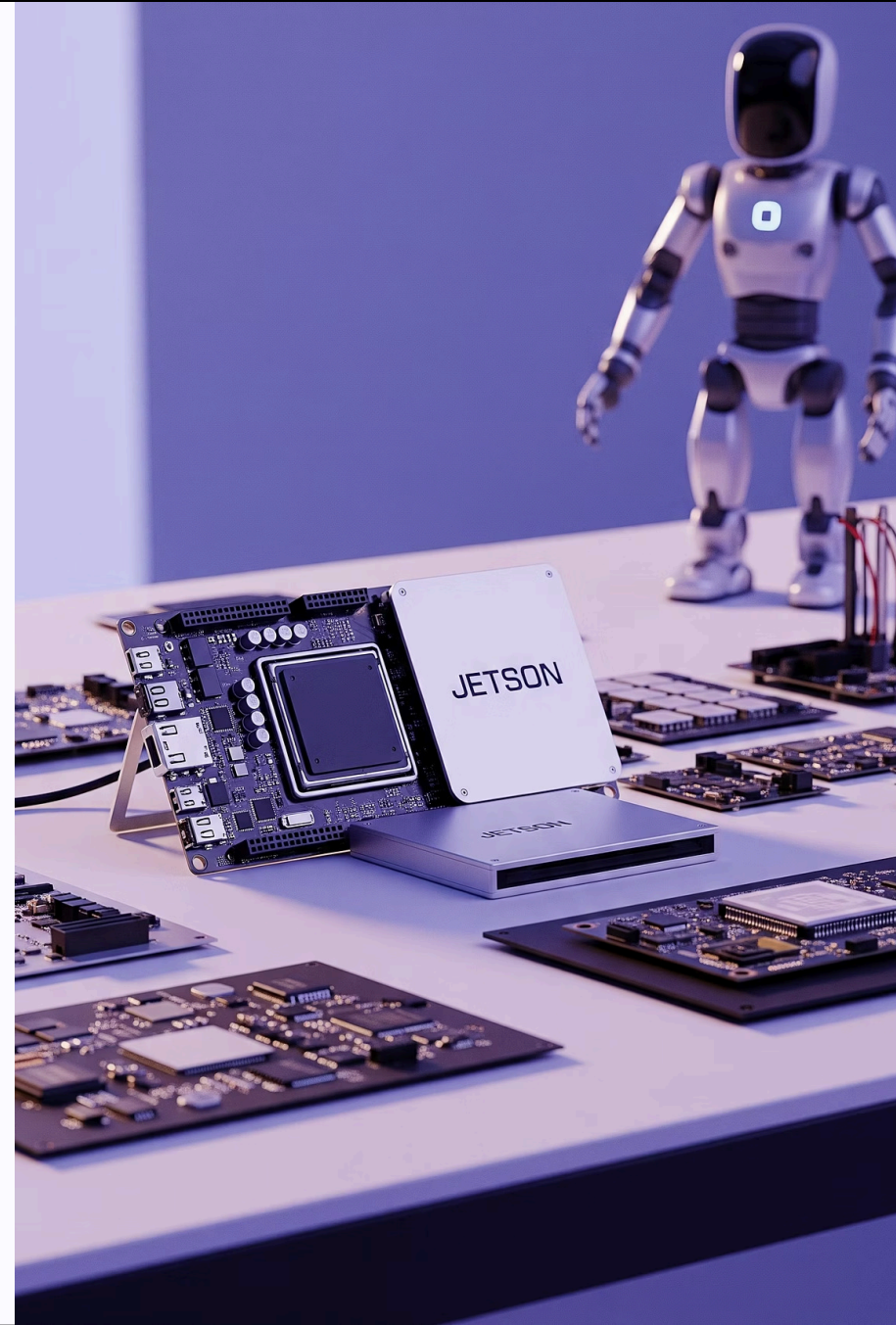


# Introduction to NVIDIA Jetson Devices

## AI at the Edge, Powered by NVIDIA

Bringing powerful GPU-accelerated AI computing to embedded systems and IoT devices, enabling real-time inference where data is generated.



# What Is NVIDIA Jetson?

NVIDIA's **edge AI hardware platform** that combines **ARM CPUs + CUDA-capable GPUs** on a compact module, delivering supercomputer performance in an embedded form factor.

Optimized specifically for **vision applications, robotics, and autonomous systems** that require real-time AI processing at the edge.



Supports popular AI frameworks including PyTorch, TensorFlow, and ONNX, enabling seamless deployment of trained models.

# Why Jetson for Edge AI?



## GPU Acceleration

Enables real-time inference at the edge with dedicated CUDA cores optimized for parallel processing



## Compact & Power-Efficient

Small form factor and low power consumption fits perfectly in embedded and IoT applications



## Scalable Family

Modules range from hobby projects to enterprise-grade robotics applications



## JetPack SDK

Comprehensive package with prebuilt drivers, CUDA, TensorRT, and cuDNN libraries

# Jetson Family Overview



## Nano

Entry-level platform for AI hobbyists and prototypes

Perfect for learning and development



## TX2

Mid-tier solution for vision applications and drones

Balance of performance and power efficiency



## Xavier NX

Powerful edge inference in a compact package

Ideal for professional edge deployments



## AGX Xavier

Enterprise/robotics-grade computing

Advanced autonomous capabilities



## Orin Series

Latest generation with exceptional TOPS performance

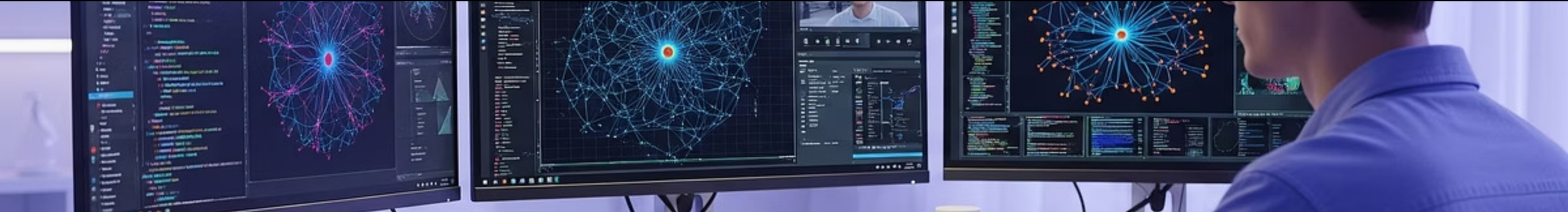
Industry-leading AI computing

# Typical Specs Across the Jetson Family

| Feature           | Entry (Nano) | Mid-Range (NX) | High-End (AGX Orin) |
|-------------------|--------------|----------------|---------------------|
| CUDA Cores        | 128          | 384-512        | 2,048+              |
| Memory            | 2-4GB        | 8-16GB         | 32-64GB LPDDR5      |
| AI Performance    | 0.5 TFLOPS   | 21 TOPS        | 275 TOPS            |
| Power Consumption | 5-10W        | 10-20W         | 15-60W              |
| Form Factor       | 69.6 x 45 mm | 70 x 45 mm     | 100 x 87 mm         |

The Jetson family scales from entry-level to high-performance computing, with up to a 550x increase in AI performance across the product line.

Power efficiency remains a priority across all tiers, with performance-per-watt increasing dramatically in newer generations.



# Jetson Software Ecosystem



## JetPack SDK

Complete OS, drivers, CUDA toolkit, cuDNN, and TensorRT for optimized inference  
Provides the foundation for all Jetson development



## DeepStream SDK

Optimized video analytics pipeline framework for multi-stream processing  
Enables efficient camera and sensor data processing at scale



## TAO Toolkit

Transfer learning and model optimization framework for domain adaptation  
Fine-tune pre-trained models with custom data



## Isaac SDK

Robotics framework for simulation, navigation, and perception  
Accelerates development of autonomous machines

# Edge AI Use Cases with Jetson

- **Computer Vision**

Real-time object detection, defect inspection, and quality control in manufacturing environments

High-throughput video analytics for multiple camera streams

- **Robotics**

Autonomous navigation, simultaneous localization and mapping (SLAM), and robotic manipulation

Multi-sensor fusion for environmental perception

- **Healthcare**

Portable imaging devices, diagnostic assistants, and patient monitoring systems

Privacy-preserving local processing of sensitive medical data

- **Retail/Smart Cities**

Intelligent surveillance, customer analytics, and personalized shopping experiences

Traffic monitoring and crowd management



# Why Jetson vs Alternatives



## GPU Acceleration

Dramatically outperforms CPU-only edge boards for parallel workloads and neural network inference

## CUDA/TensorRT Ecosystem

Complete software stack optimized for AI performance, with hardware-accelerated libraries

## NVIDIA Community & Support

Extensive documentation, forums, and enterprise-grade support options

## Scalability

Same development stack works across all Jetson modules, from Nano to Orin

# Deployment at Scale

## Hardware Integration

Jetson modules integrate easily into custom carrier boards and OEM designs

Standard interfaces and pinouts enable scalable production

## Fleet Management

NVIDIA Fleet Command and Kubernetes support for edge device orchestration

Monitor health, performance, and manage workloads remotely

## Software Updates

Over-the-air (OTA) updates for models, applications, and system software

Containerized deployment for reproducible application environments

## Security

Secure boot and encrypted storage protect sensitive AI models and data

Hardware root of trust ensures system integrity



# Key Takeaways

**Jetson** = **edge AI powerhouse** in compact form factors with industry-leading GPU acceleration

**Flexible product range** from Nano to Orin supporting any workload scale and power envelope

**Comprehensive ecosystem** with JetPack, DeepStream, TAO, and Isaac SDKs for rapid development

**Ideal platform for vision, robotics, autonomous systems**, and any application requiring real-time AI inference



Questions?